

Multiplication and division of monomials

Numbers with numbers, letters with letters — and the laws of indices do the rest.

LESSON 66–67

2 × 40 MIN

MATEMATIKA 7, §30

S8 2.1

LESSON CLOCK

15'

25'

25'

20'

5'

Multiplying	15 min
Raising to a power	25 min
Dividing	25 min
Areas and volumes	20 min
Homework	5 min

LEARNING OBJECTIVES

- Multiply two or more monomials and give the answer in standard form.
- Divide one monomial by another when the division is exact.
- Raise a monomial to a power.
- Use the operations in area and volume problems.

TEXTBOOK

National	pp. 182–188
Cambridge	Stage 8, pp. 12–18
Workbook	Exercise 2.1

01

Explanation

Multiplying

Multiply the coefficients; then, for each letter, add the exponents.

PRODUCT	NUMBERS	LETTERS	ANSWER
$3a \cdot 4b$	12	ab	$12ab$
$2x^2 \cdot 5x^3$	10	x^5	$10x^5$
$(-3a^2b)(4ab^3)$	-12	a^3b^4	$-12a^3b^4$
$(\frac{1}{2}xy)(6x^2)$	3	x^3y	$3x^3y$

THE PRODUCT OF TWO MONOMIALS IS ALWAYS A MONOMIAL

Its degree is the sum of the two degrees. That is worth checking as you go: $(-3a^2b)$ has degree 3, $(4ab^3)$ degree 4, and the answer degree 7 ✓

Raising to a power

$$(ka^mb^n)^p = k^pa^{mp}b^{np}$$

Every factor is raised — the coefficient included.

EXPRESSION	ANSWER
$(2x)^3$	$8x^3$
$(3a^2b)^2$	$9a^4b^2$
$(-2x^3)^3$	$-8x^9$
$(-2x^3)^2$	$4x^6$

THE COEFFICIENT IS RAISED TOO

$(3a^2b)^2 = 9a^4b^2$, not $3a^4b^2$. Forgetting to square the 3 is the standard slip.

Dividing

Divide the coefficients; then, for each letter, subtract the exponents.

QUOTIENT	NUMBERS	LETTERS	ANSWER
$12a^5 \div 4a^2$	3	a^3	$3a^3$
$-15x^3y^2 \div 5xy$	-3	x^2y	$-3x^2y$
$8a^4b^3 \div (-2a^4b)$	-4	b^2	$-4b^2$

THE QUOTIENT IS A MONOMIAL ONLY IF EVERY EXPONENT STAYS NATURAL

$a^2 \div a^5$ is not a monomial. In Grade 7 the divisions set are always exact, but the condition is worth knowing.

Areas and volumes

FIGURE	DIMENSIONS	AREA OR VOLUME
rectangle	$3a$	$12ab$
square	$2x$	$4x^2$
cuboid	$a, 2a, 3a$	$6a^3$
cube	$2x$	$8x^3$

The first row assumes a rectangle $3a$ by $4b$. Notice that the units follow the degree: a length is degree 1, an area degree 2, a volume degree 3.

THE DEGREE IS A DIMENSION CHECK

An answer of degree 2 for a volume is wrong before any arithmetic is examined. This is the algebraic version of checking the units.

02

Worked examples

EXAMPLE 1 Simplify $(-3a^2b)(4ab^3)$.

1

Numbers: $-3 \times 4 = -12$.

2

a : $a^2 \cdot a = a^3$.

3

b : $b \cdot b^3 = b^4$.

4

$= -12a^3b^4$
Degree 7 ✓

ANSWER

$-12a^3b^4$

EXAMPLE 2 Simplify $(3a^2b)^2$ and $(-2x^3)^3$.

① Square every factor: $3^2 = 9$, $(a^2)^2 = a^4$, b^2 .

② $= 9a^4b^2$

③ Cube every factor: $(-2)^3 = -8$, $(x^3)^3 = x^9$.

④ $= -8x^9$

ANSWER $9a^4b^2$ and $-8x^9$ **EXAMPLE 3** Simplify $-15x^3y^2 \div 5xy$.

① Numbers: $-15 \div 5 = -3$.

② x : $x^3 \div x = x^2$.

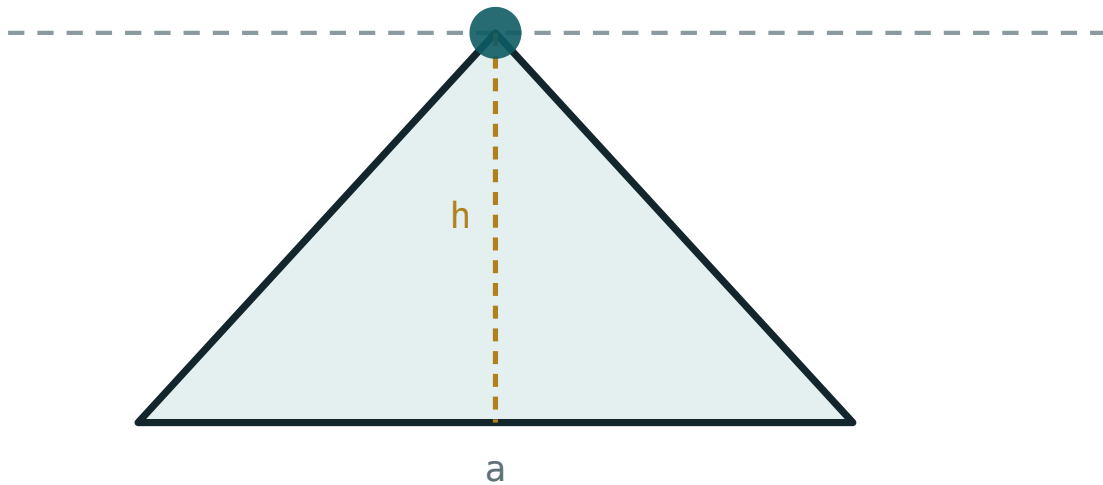
③ y : $y^2 \div y = y$.

④ $= -3x^2y$

ANSWER $-3x^2y$ **03 Interactive model**

Draw a rectangle $3a$ by $4b$ on the board and ask for its area before any rule is stated; the class multiplies numbers and letters separately without being told to.

A rectangle with algebraic sides

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To multiply	Ko'paytirish	Умножить
To divide	Bo'lish	Разделить
Standard form	Standart ko'rinish	Стандартный вид
Exactly divisible	Butun bo'linadi	Делится нацело
Power of a monomial	Bir had darajasi	Степень одночлена
Area	Yuza	Площадь
Volume	Hajm	Объём
Cancel	Qisqartirish	Сократить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$2x^2 \cdot 5x^3$ equals:

$(3a^2b)^2$ equals:

$(-2x^3)^3$ equals:

$12a^5 \div 4a^2$ equals:

The degree of a product of monomials is:

A cube of side $2x$ has volume:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $3a \cdot 4b$

ANSWER $12ab$

2. $2x^2 \cdot 5x^3$

ANSWER $10x^5$

3. $(2x)^3$

ANSWER $8x^3$

4. $(3a^2b)^2$

ANSWER $9a^4b^2$

5. $12a^5 \div 4a^2$

ANSWER $3a^3$

6. $(-2x^3)^2$

ANSWER $4x^6$

7. $(-2x^3)^3$

ANSWER $-8x^9$ **Medium · the standard question**

7 PROBLEMS

1. $(-3a^2b)(4ab^3)$

ANSWER $-12a^3b^4$

2. $(\frac{1}{2}xy)(6x^2)$

ANSWER $3x^3y$

3. $-15x^3y^2 \div 5xy$

ANSWER $-3x^2y$

4. $8a^4b^3 \div (-2a^4b)$

ANSWER $-4b^2$

5. Area of a rectangle $3a$ by $4b$

ANSWER $12ab$

6. Volume of a cuboid a , $2a$, $3a$

ANSWER $6a^3$

7. Degree of $(-3a^2b)(4ab^3)$

ANSWER 7

Hard · stretch and reason

7 PROBLEMS

1. $(2a^2b)^3 \cdot 3ab$

ANSWER $24a^7b^4$

2. $\frac{(3xy^2)^3}{9x^2y^4}$

ANSWER $3xy^2$

3. $(-a^2b)^4 \div (a^3b^2)$

ANSWER a^5b^2

4. A cube of volume $27x^6$: its edge

ANSWER $3x^2$

5. A square of area $49a^4$: its side

ANSWER $7a^2$

6. $(2x)^2 \cdot (3x)^3$

ANSWER $108x^5$

7. Why is $a^2 \div a^5$ not a monomial?

ANSWER The exponent would be negative

07 Homework**Homework — 5 tasks**

Check the degree of every answer against the degrees you started with.

1. Simplify $(-2a^3b)(5ab^2)$.
2. Simplify $(4x^2y)^2$ and $(-3a)^3$.
3. Simplify $20a^6b^3 \div 5a^2b$.
4. Find the volume of a cuboid with edges $2a$, $3a$ and $4b$.

5. A cube has volume $64x^3$. Find its edge.